Instructions

- Submit the assignments in time via online form here.
- The solutions MUST be done in LaTeXand handed as pdf. No handwritten assignments will be accepted.
- For python based problems, provide the codes as well as any generated figure. Seek your instructor's or peers' help if you need it. Responses during the viva that amount to: '...that's what the AI gave me', would result in a 0.
- Equations must be numbered and properly formatted.
- Expect some simple questions from the readings/viewings in the exam as well.
- Your final marks in the assignment will be contingent on the viva.

LaTeXTips

If you feel your LaTeXfile is getting unwieldy, you can use the `input` command to divide it into smaller parts. See this.

Vector Calculus

1. Solve problem 1.57 of Griffiths.

Electrostatics

2. Griffiths Problem 3.27

Magnetostatics

3. Consider a long, dielectric right circular cylinder of radius of R_0. The cylinder has a radial polarization of, $P = 0.5P_0 r\hat{r}$, where r is the radial distance from the axis of the cylinder.

   a) Find the charge density ρ in the cylinder and the surface charge density σ.

   b) If the current rotates with angular velocity ω, find the volume current density j and the surface current density K.

   c) Now find the magnetic field due to these surface and volume currents.

Reading/Writing

Write a 700 word essay on altermagnetic materials. Your essay must provide at least 10 bibliographic references from the last ten years.

Note that the citation style must follow the American Institute of Physics (AIP) guidelines.
To see how to insert references in LaTeX, see this.